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Novel process parameters based approach for reducing residual stresses in WAAM

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ABSTRACT

Residual stresses are unfavourable phenomena in wire arc additive manufacturing, which needs to be critically analysed and mitigated. Process parameters play a crucial role in mitigating stress, the work here attempts to analyse strategies to find the best trend of varying the process parameters rather than selecting the optimal process parameters in order to have the least residual stresses. The initial part of work deals with optimising the input parameters assuming uniform value of parameters to build the entire structure wherein local optimal values of current, voltage and welding speed are obtained. However, critically analysing for global optimas, the parameters needed further reduction, which would create nodus of arc stabilities. To address this an innovative procedure is applied wherein a variation trend of parameters along the length of the build structure are used and trend analysis of the output is carried out to find the best and worst case. The study's outputs indicate that the difference in the effective stress formed via designed ideology and the one with uniform parameters on an average scale is 210 MPa.
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1. Introduction

Wire arc additive manufacturing(WAAM) is one of the most widely used additive manufacturing techniques due to its ability to produce parts at an extremely fast rate, nearly about 10 kg/hr [1]. The WAAM technique works on the combined principles of welding and traditional additive manufacturing; where feedstock in form of filler wire is melted and deposited in a form that resemblance individual layer when on a cumulative scale would comprehend the final product. The process of WAAM holds its supremacy in terms of cost of manufacturing, the raw material used is almost 1/10 of the cost that is incurred to that of the powder-based process [2] moreover the cost of working is also drastically less. The layers as being build indulge a greater amount of thermal input than most other additive manufacturing. This thermal input creates a gradient of temperature variation, which causes defects [3]. The primary defects caused on the parts due to this temperature gradient, cooling rate and excessive energy input are distortion and residual stress [4]. Each layer is being acted upon by the varying rate of heating and cooling, which causes differential rates of expansion and contraction in the build structure, causing it to deform and develop residual values of stress.

Literatures over the year have tried to modify the effect of residual stress by multiple means. A few notable means are carrying out an in-situ rolling in top and on the side of built structure at end of process [5] and as an interpass process [6]. Methods of heat treatment along with laser peening [7], using a proper path planning strategy [8] have also proven to reduce the residual stress in WAAM. However, limited literatures are availed on parametrical variation for reducing the as-build residual stresses. Based on literatures [9,10] it could be inferred that the modification of the input parameters alike current, voltage and the travel speed could possible contribute variation of the residual stress and due to limited works available forehand it was decided to be studied upon.

As known by literature [8,11] that the residual stresses are a function of the heat input, which is inevitable in welding but by analysis the physics of the welding, it could be possibly stated that by properly modifying the heat input into the system [12,13] the values of residual stress could be constraint considerably. In welding, the heat input is a function of the input parameters like current, voltage and welding speed so by selecting proper value of

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Table 1
Details of build wall.

Substrate	300mm × 80mm × 20mm
Weld bead size	100mm × 5mm × 2.2 mm
Ambient Temperature	303 K
Gravity constant	9.80665 m/s^2 (-Z direction)
Constrain of base plate-SIMUFACT	Bearing (300mm × 80mm × 20mm)
Circular Fixings	Dia-15 mm, Height-5 mm to arrest distortion [16]
Interpass timing	20sec
Number of layers	10

such parameters the residual stress of the system could be considerably modified.

Therefore, to validate the physics behind the heat input and its relation to initial trial simulation, substantiate the same. The work on initial simulation trial is done on SIMUFACT WELDING, a Finite Element software dedicated for simulation of welding processes and WAAM. Only a few number of literatures have dealt which use the software of SIMUFACT WELDING, wherein WAAM and welding are done using the simulation software and effective stresses, which is a measure of residual stress, is studied. Works by Sun et al., [14] have tried out to investigate systematically the residual stress of the component produced by WAAM process, for which a set of numerical analyses had been carried out by using Simufact Welding software. Huang et al., [15] in his works had studied the effect of post-weld heat treatment on the deformation and the residual stresses of dissimilar welding of 20Cr18Ni9 by using simulations with the aid of Simufact welding. Therefore, the outset of the work was to simulate WAAM of a part to find out the best possible way of depositing to have the least values of residual stress.

2. Simulation details

For simulation of the WAAM in SIMUFACT WELDING, the details of the input are shown Table 1.

The geometry is constructed in Abaqus and the type of pass used is similar in all the cases and are as shown in Fig. 1. The entire height of the build wall is divided into 11 layers and chemical composition of material is as shown in Table 2.

2.1. Heat source modelling

The heat source is modelled based on Goldak's double ellipsoidal equation [17], where the heat source is double-ellipsoidal and is ascertained to carry the input energy throughout the entire surface of the WAAM component build. Fig. 2 represents Goldak's Model [18]. The rear and the front parts are the parts represents areas to carry heat source. The equation of the heat sources for the front and the rear end are:.

Front:.

$$q_f(x,y,z) = \frac{6\sqrt{3}.f_f.Q}{a.c_f.b.\pi\sqrt{\pi}}.e^{\left(-3x^2/a^2\right)}.e^{\left(-3y^2/b^2\right)}.e^{\left(-3z^2/{c_f}^2\right)} \quad (1)$$

Rear:.

$$q_r(x,y,z) = \frac{6\sqrt{3}.f_r.Q}{a.c_r.b.\pi\sqrt{\pi}}.e^{\left(-3x^2/a^2\right)}.e^{\left(-3y^2/b^2\right)}.e^{\left(-3z^2/{c_r}^2\right)} \quad (2)$$

and.

$$f_f + f_r = 2, Q = \eta VI \quad (3)$$

where a,b is the half-width and depth of the elliptical model of heat source and c_f,c_r are lengths of front and rear elliptic respectively. Parameters f_f and f_r represent distribution factor for the temperatures in the front and rear sides respectively. Q represents the heat input into the system where η represents the efficiency, V and I represents the voltage and current respectively.

So based on the literature [9] and trials done the values of the Goldak's parameters are as shown in Table 3.

Convection and radiation are the assumed modes to transfer the heat into the environment from all the plane other than those of the symmetrical ones [19] and are calculated by the following set of equations [20].

$$q_{bc} = h(T - T_a) + \varepsilon\sigma\left(T^4 - T_a^4\right) \quad (4)$$

$$h_{total} = h + \varepsilon\sigma(T + T_{amb}).\left(T^2 + T_{amb}^2\right) \quad (5)$$

where,h is convection coefficient,ε is radiation factor of a material, q_{bc} is waste heat, σ is Boltzmann constant, T_{amb} is environment temperature and T is the material temperature.

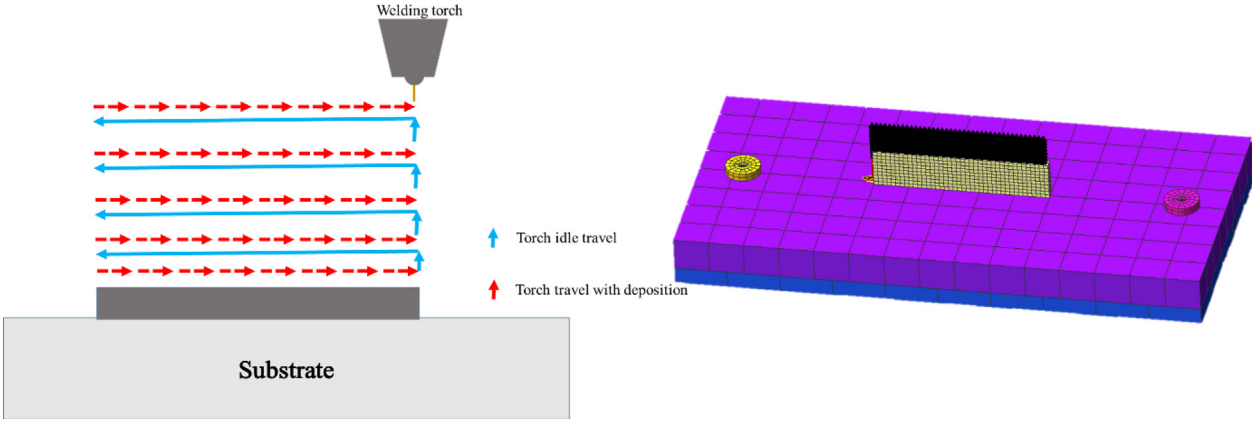


Fig. 1. (a) Torch travel path; (b) Basic build structure of WAAM.

Table 2
Chemical composition of alloy (wt%).

C	Si	Mn	P	S	Cr
0.14–0.19	≤0.40	1.00–1.30	≤0.025	≤0.035	0.80–1.10

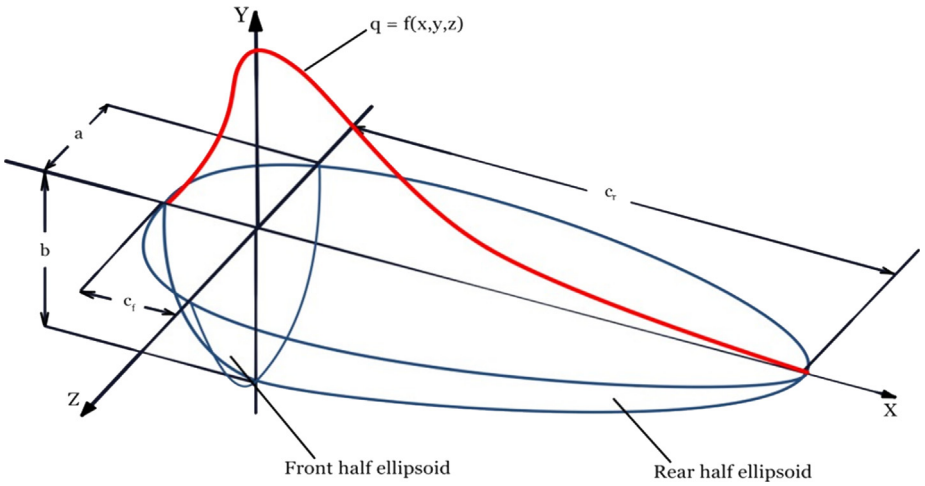


Fig. 2. Double ellipsoidal model.

Table 3
Parameters for Goldak's model.

Parameter	Value(mm)
Front length(a_f)	2
Rear length(a_r)	7
Width(b)	2.5
Depth(d)	2.3
Heat front scaling factor	0.444
Efficiency	0.60

The mechanical analysis of the build structure was calculated from the thermal results in terms of a heat load [14] Mechanical analysis and thermal analysis use the same Finite Element model. The interfacing temperature-dependent mechanical properties are taken in from the material database of the SIMUFACT WELDING software. The effect of the strain hardening was totally ignored because of strain-softening that is being introduced by annealing effect [14].

For mechanical analysis, the results of thermal analysis are used as input data for the respective nodes. The elasto-plastic strains for the respective analysis is calculated as [21].

$$\varepsilon_{ij} = \varepsilon_{ij}^{e} + \varepsilon_{ij}^{p} + \varepsilon_{ij}^{th} \tag{6}$$

Where, ε_{ij} is the total strain,ε_{ij}^{p} is the plastic strain, ε_{ij}^{e} is the elastic strain and ε_{ij}^{th} is thermal stress and for an isotropic equation, the stress can be calculated by the following equation.

$$\sigma_{ij} = C_{ijkl}^{e}(\varepsilon_{ij} - \varepsilon_{ij}^{p} - \varepsilon_{ij}^{th}) \tag{7}$$

Where,C_{ijkl}^{e} is the elastic stiffness matrix of the particular material.

2.2. Range of parameters for simulation

Through trial experimentation and based on visual inspection for single layer bead the range of parameters was decided on, out of which three values at stipulated intervals (Table 4) are selected to reduce the number of iterations and simulation time.

To analyse the wholesome effect of all the parameters a complete raster analysis is done which requires 27 experiments, whose combinations are tabulated in Table 5.

Table 4
Range of parameters.

	Level-1	Level-2	Level-3
Current(amps)	80	100	120
Voltage(volts)	10	14	18
Welding speed(mm/min)	60	70	80

3. Results and discussion

The simulation is done on SIMUFACT welding. The simulations for the orthogonal build array are done on the assumption that ambient conditions and the build conditions are similar in all cases. The initial motive was to study whether optimising parameters could reduce the residual stresses and study the most optimal mix of parameters to obtain least values of residual stresses. Residual stress is an unfavourable phenomenon [5] in WAAM components, so inorder to study this the values of the maximum residual stresses are taken for analysis.

Table 5
Residual stresses.

Current(A)	Voltage(V)	Speed(mm/min)	Residual stresses(MPa)
80	10	60	345.4
80	10	70	353.0
80	10	80	360.9
80	14	60	372.8
80	14	70	366.9
80	14	80	355.6
80	18	60	372.2
80	18	70	351.4
80	18	80	375.5
100	10	60	361.3
100	10	70	349.1
100	10	80	355.6
100	14	60	356.6
100	14	70	358.0
100	14	80	369.9
100	18	60	367.3
100	18	70	376.6
100	18	80	376.0
120	10	60	369.8
120	10	70	369.1
120	10	80	381.1
120	14	60	369.0
120	14	70	375.9
120	14	80	370.6
120	18	60	394.7
120	18	70	358.6
120	18	80	360.5

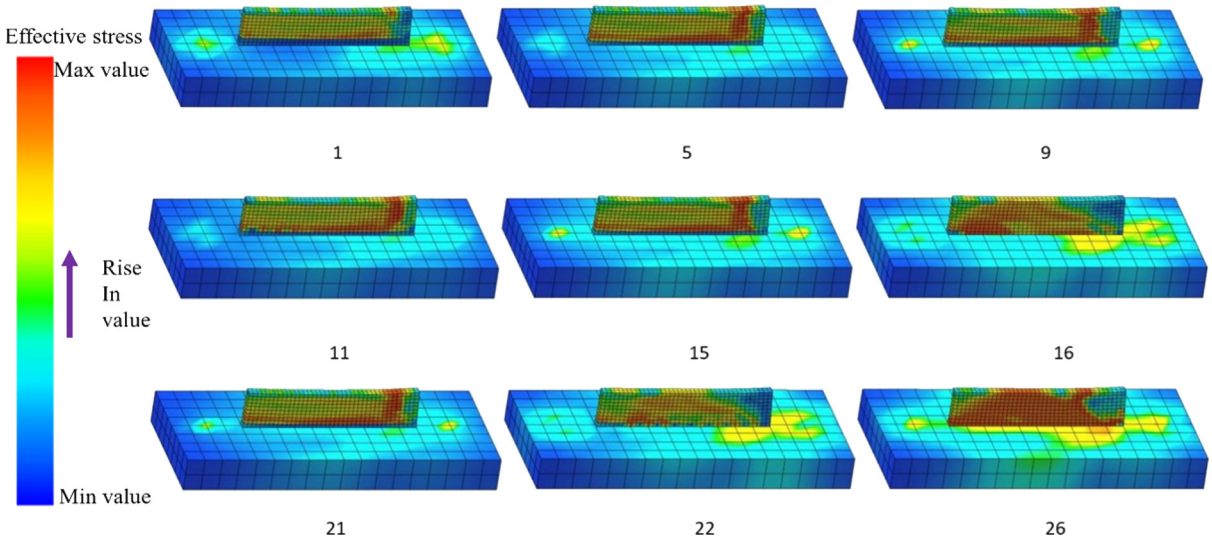


Fig. 3. Simulation of build wall structure.

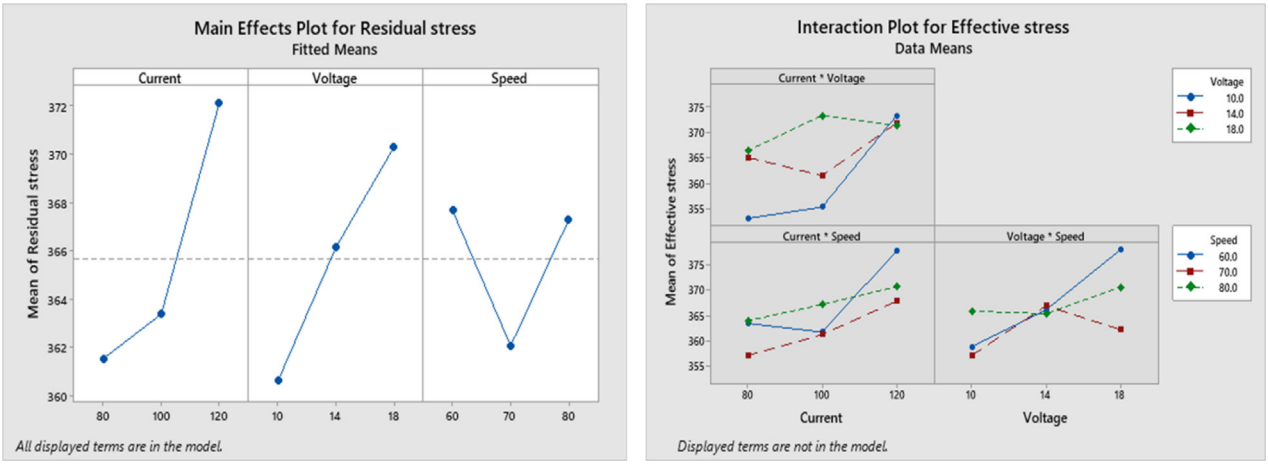


Fig. 4. Main effect and interaction plots for residual stress.

3.1. Simulation of WAAM

The simulation has been carried out for the given set of parameter and results obtained are tabulated in Table 5 and samples are shown in Fig. 3.

After the simulation is carried out an S/N ratio analysis is carried out to test the effect of noise on the output, followed by ANOVA where contributions by each welding process parameter is studied from where it could be understood that the current and voltage hold significant importance over the welding speed.

3.2. Effect of various parameters

After ANOVA analysis is completed, the optimisation for the L27 array is done. The plot of the effect of various parameters on residual stress is as shown in the Fig. 4. And the 3D plot of the effect of parameters are jotted as shown in Fig. 5.

From the above figures, it could be said that the local optima for the current and voltage are 80amps and 10 voltage and for welding speed are 70 mm/min. However, precisely speaking (Fig. 4) the optimised value is obtained for the welding speed alone, as the values of current and voltage tend to show off a decreasing trend, which possibly brings out the fact that a further reduction of the current and voltage would result in a better value of residual stress. But reducing the values even further can possibly extinguish the arc and may cause the bead to be irregular [22]. Therefore, it would be erroneous to conclude that these are the optimal parameters. After a brief brainstorming session, the next level of study is decided to be done by a variation change in the parameters.

3.3. Trend based variations of parameters

This study aims to reveal the best trend of parameters variation as the component is being built so that part with least values of residual stresses is obtained. The study assume quiet element method [19] so it is assumed that the variation of the parameter doesn't significantly affect the bead geometry and as a one on one relationship between the parameters and bead geometry would not be obtained at ease the assumption of no effect of parameters are presumed. After thorough rasterisation and trial simulation, the set of parametrical variation as in Table 6 is decided to be followed upon.

There has been a specification of a trend of increasing and decreasing which mean the trends of variation of the parameters are on a linear scale as in Table 7 and the middle layer is at a constant value as specified (except for Voltage variation case 4- as

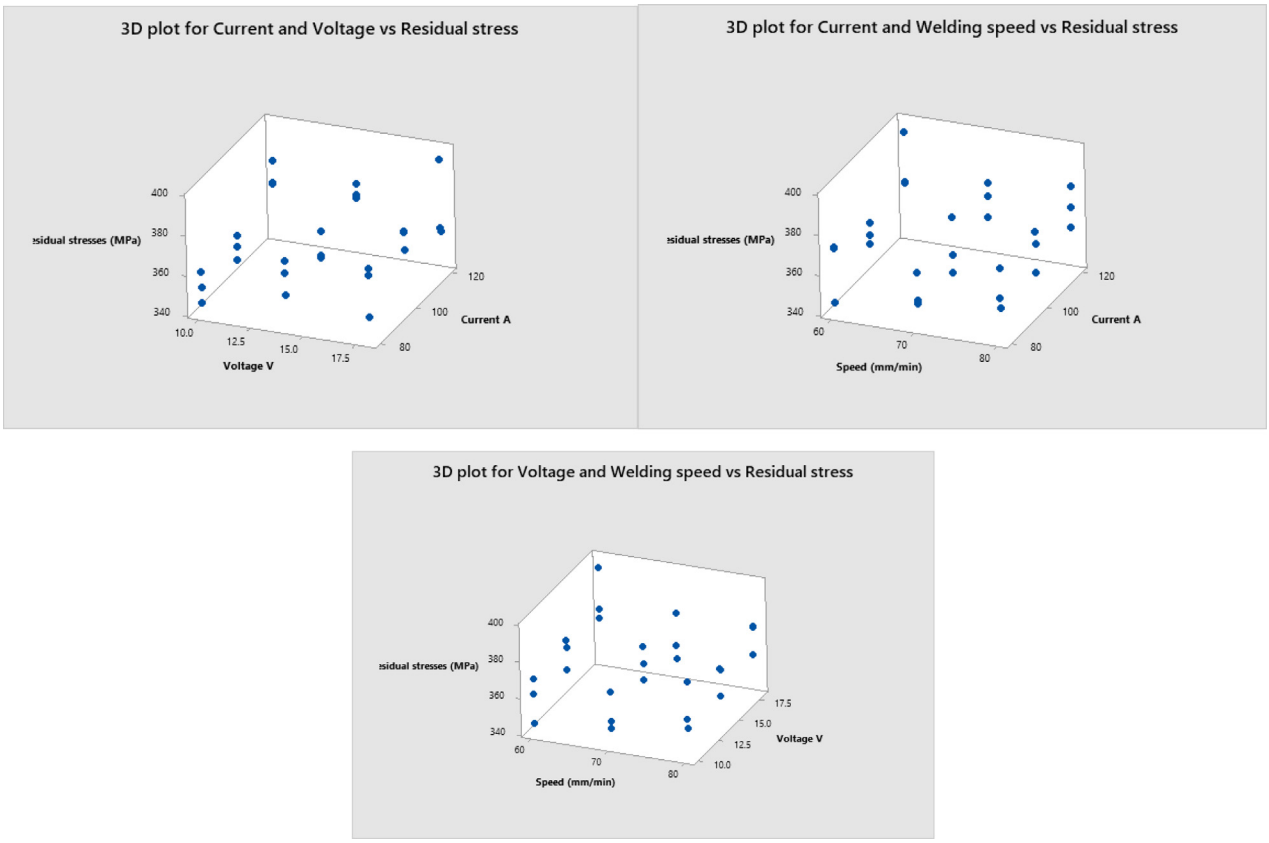


Fig. 5. Box plot for relation between various parameters and residual stress.

reduction would possible extinguish). The trails simulations are carried out on SIMUFACT welding and the results are as shown in the Table 6.

So based on the output data, trend analysis is done to study relationship between the parametrical variation and the output residual stress based on which graphs are plotted (Fig. 6).

From the Fig. 6, we can infer that on average, when the welding speed decreases and the voltage increases incrementally, the effective stress reduces. The current graph is not unidirectional within the given base settings so a conclusive trend cannot be inferred so the current setting of constant is assumed for further analysis. Therefore, the best case is that of increasing voltage, decreasing welding speed and constant current and the worst case is that of decreasing voltage, increasing welding speed and constant current.

Table 6
Parameters variation along the length.

Current(A)	Voltage(V)	Speed(mm/min)	Stress(MPa)
80	10	60	345.4
80	10	Inc-(Middle-60)	358.9
80	10	Dec-(Middle-60)	323.4
80	Inc-(Lowest-10)	60	324.6
80	Dec-(Middle-10)	60	387.3
Inc-(Middle-80)	10	60	344.1
Dec-(Middle-80)	10	60	354.0
100	14	70	358.3
100	14	Inc-(Middle-70)	356.2
100	14	Dec-(Middle-70)	372.5
100	Inc-(Middle-14)	70	353.1
100	Dec-(Middle-14)	70	368.1
Inc-(Middle-100)	14	70	374.8
Dec-(Middle-100)	14	70	370.9
120	18	80	399.6
120	18	Inc-(Middle-80)	386.5
120	18	Dec-(Middle-80)	364.3
120	Inc-(Middle-18)	80	370.1
120	Dec-(Middle-18)	80	397.5
Inc-(Middle-120)	18	80	367.8
Dec-(Middle-120)	18	80	393.7

Inc*-Increasing, Dec*-Decreasing.

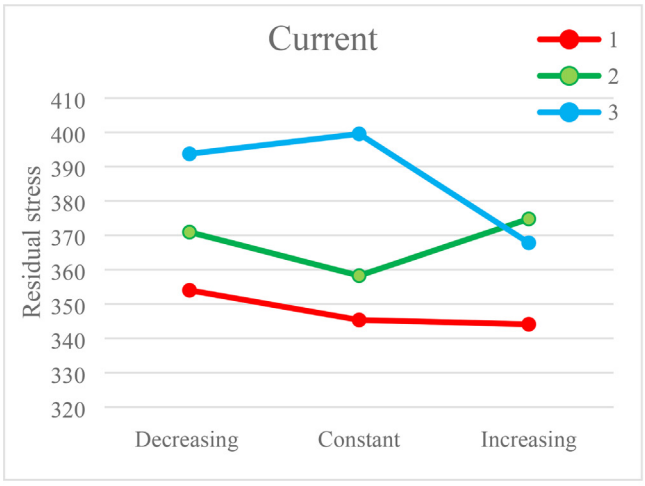
Table 7
Range of variation.

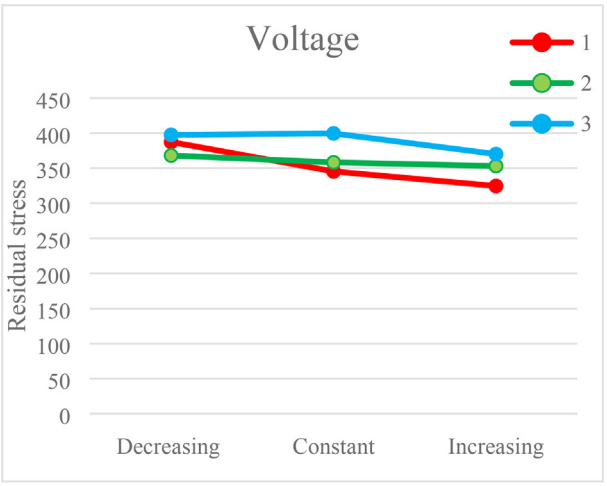
	Current (Amps)	Voltage (Volt)	Welding speed(mm/min)
Increasing(per layer)	5	0.5	5
Decreasing(per layer)	5	0.5	5

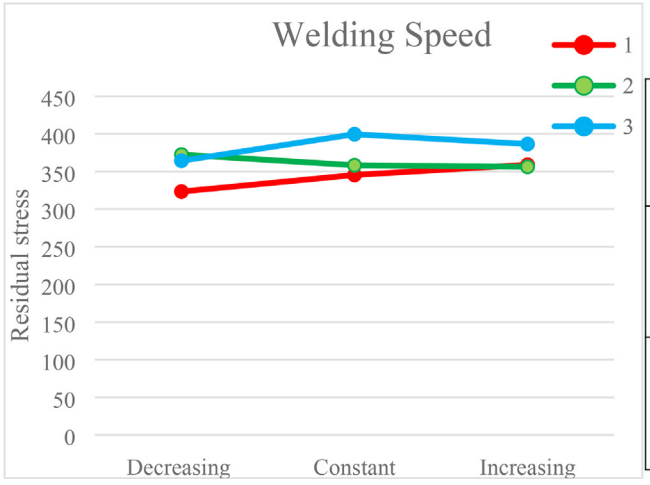
3.4. Validation:

To validate the results, the combined effect of the parameters for the same geometry the best case and the worst case are compared. The results show that the best case has maximum effective stress of 330.34 MPa and the worst case has 387.86 MPa. A significant difference of 57.52 MPa is observed as shown in Fig. 7.

For re-validation of the innovative idea, a different geometry is considered. The best case and worst case are simulated and the results are shown in Fig. 8 which also tend to follow the same pattern. The maximum effective stress of the best case is 586.80 MPa, the maximum effective stress of the worst case is 1086.7 MPa, the average value of difference in the residual stresses for the innovative ideology, and the optimal one from previous case is of 173 MPa. Thereby validating the innovative way of modifying the stress profile.







		Decreasing	Constant	Increasing
Speed	1	323.372	345.377	358.916
	2	372.492	358.284	356.218
	3	364.304	399.558	386.478
		Decreasing	Constant	Increasing
Voltage	1	387.32	345.377	324.586
	2	368.075	358.284	353.092
	3	397.486	399.558	370.106
		Decreasing	Constant	Increasing
Current	1	354.024	345.377	344.123
	2	370.939	358.284	374.78
	3	393.739	399.558	367.81

Fig. 6. Trend analysis.

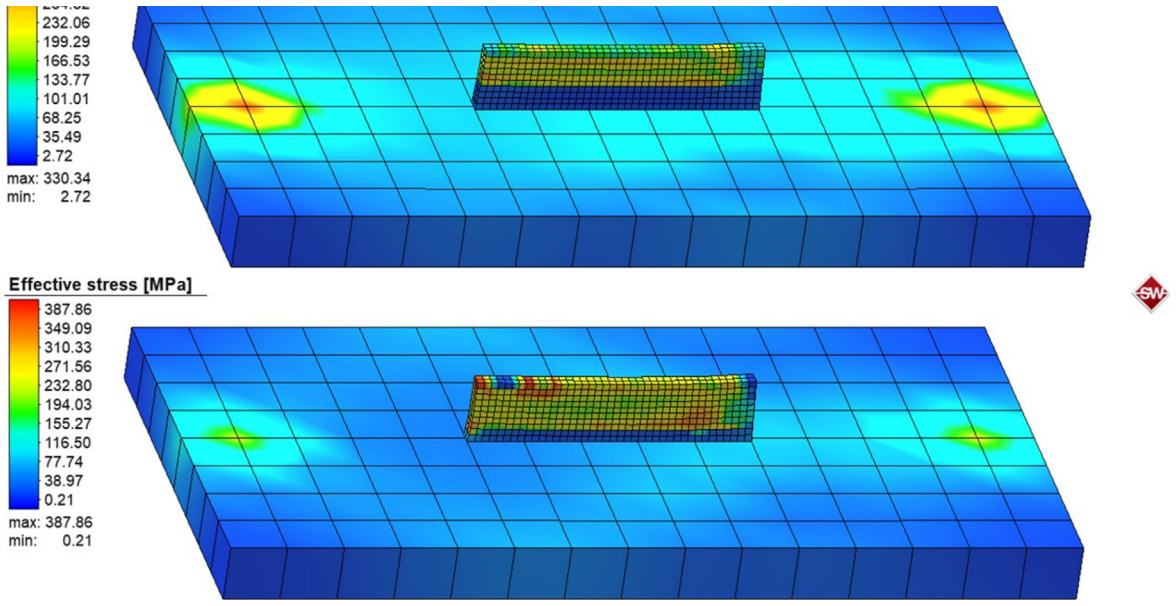


Fig. 7. Validation of result on same geometry.

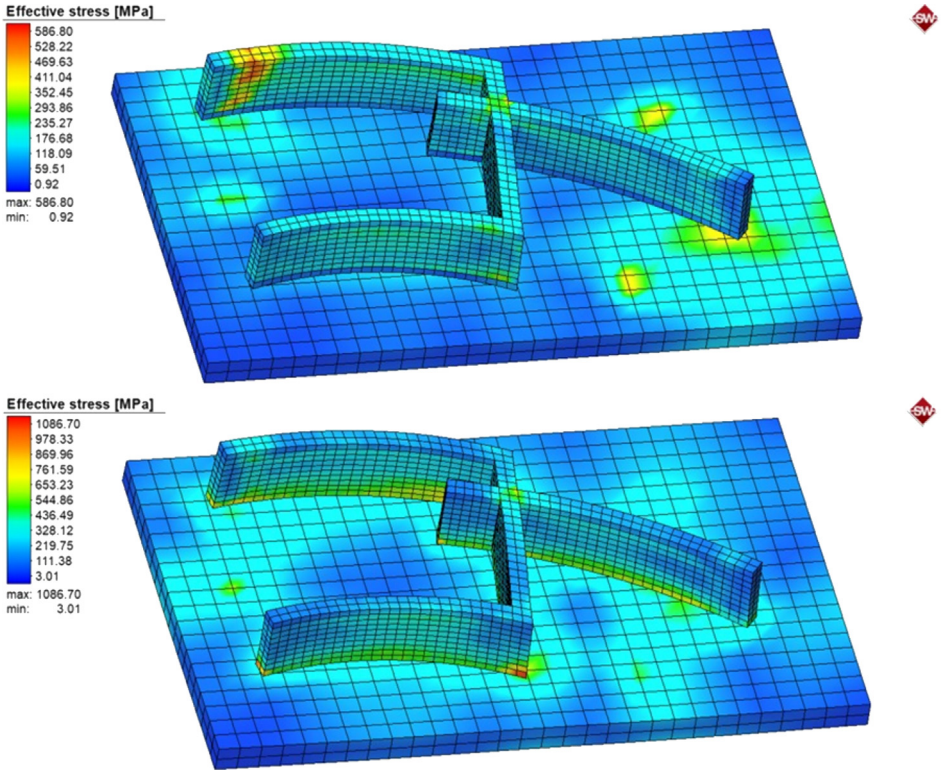


Fig. 8. Validation of result on same geometry (a) Best case and (b) Worst case.

4. Conclusion

The work attempts to understand the variation of process parameters on the as build residual stresses. The initial part of work attempts the optimizing the residual stresses by considering uniform set of parameters to build the entire component. The results obtained by constant parametrical change for optimization showed that the change of parameters for every layer would result in local optima for individual parameters but the changing trend shows that further of individual parameters would result in better values. The problem with further reducing the parameters would be that arc would be in terms of arc stability and build quality would be affected. Then a least explored method of having a variational trend of parametrical changes where studied upon inorder to explore the possibility of even reducing the residual stresses, in which different trend alike keeping a set of parameters constant and further changing the other parameters in a detrimental fashion was explored. Upon analysis, it could be found that this novel technique of parameters along the length of build displayed a reduction of little more than 170 MPa (for the maximum case) and on an average of 200 MPa as compared to the constant parameter variational technique, thus enabling us to build component with lesser residual stresses. This approach differs from the traditional method of optimization in sense that the variational change of parameters is distinct for a single build component rather than opposed to the traditional technique were in parametrical analysis is done for constant parameters throughout the build of the component.

CRediT authorship contribution statement

C.T. Justus Panicker: Methodology, Software. **K. Rohit Surya:** Software, Validation. **V. Senthilkumar:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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